

Qatar-1b

R-2791

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_120822 · created 2026-09-04T01:12:25+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2456161.68167 +/- 0.00015 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.188 +/- 0.055
Transit depth (Rp/Rs)^2	3.55 +/- 2.09 [%]
Orbital Inclination (inc)	80.38 +/- 2.94
Airmass coefficient 1 (a1)	4.22 +/- 1.96
Airmass coefficient 2 (a2)	-1.18 +/- 1.49
Scatter in the residuals of the lightcurve fit is	46.51 %
Best Comparison Star	#1 - [190, 234]
Optimal Aperture	2.71
Optimal Annulus	7.16
Transit Duration (day)	0.038 +/- 0.029

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.188 ± 0.055 vs 0.1463 (deviation 0.76σ)
Duration fit vs published	0.038 d vs 0.0692 d (deviation 1.07σ)
Mid-time O–C	-0.3 min
Depth SNR	3.4
Residual scatter	46.51 %

Light curve

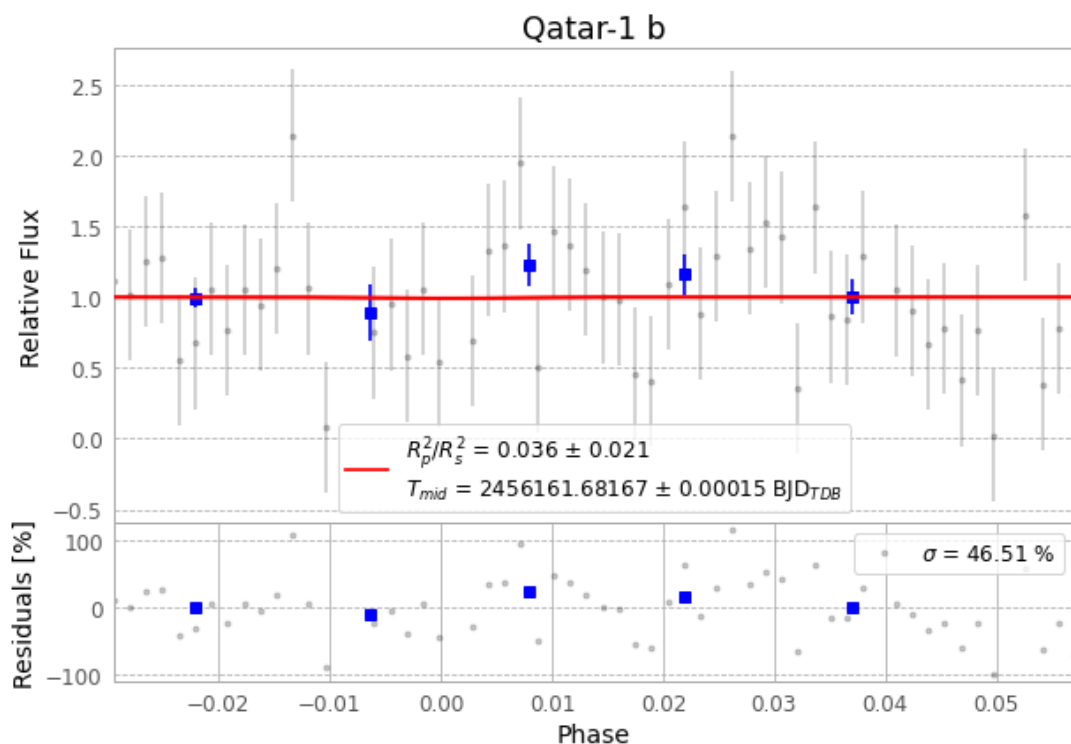


Figure 1 — FinalLightCurve_Qatar-1 b_2012-08-21.png (SHA-256 9a3259780a2546c6...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [561, 313], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 334a158097576166...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

58 FITS frames (38 MB), Qatar-1120822031813.FITS → Qatar-1120822061512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-120822.log (f2dc413b3698...), FinalLightCurve_Qatar-1 b_2012-08-21.png (9a3259780a25...), FinalLightCurve_Qatar-1 b_2012-08-21.csv (067c6271e3d9...)

Compute resources

Wall time 152.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-04 01:12Z.